

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	$A = \pi \times (0.09 \times 10^{-3})^2$ $A = 2.5 \times 10^{-8} \text{ [m}^2\text{]} (1)$ $\% \text{ uncertainty} = \frac{2 \times 0.01 \times 100}{0.18} \text{ seen or } = \frac{0.01 \times 100}{0.09} (1) \text{ accept}$ Accept first principles method leading to 10.8 % or 11.4 % or both Or 11.1 % seen	1	1		2	2	2
		(ii)	$\frac{1 \times 100}{1940} (1) \text{ ignore sig figs}$ (Or 0.05 % seen)	1			1	1	1
	(b)	(i)	Substitution: $E = \frac{9.81 \times 1.940}{2.5 \times 10^{-8} \times 6 \times 10^{-3}} (1) \text{ (ecf on A)}$ $E = 1.2688 \times 10^{11} \text{ [Pa]} (1) \text{ (ecf on slips in powers of 10)}$ $E = 127 \text{ or } 130 \text{ GPa or } 125 \text{ GPa if } 2.544 \times 10^{-8} \text{ m}^2 \text{ is used}$	1			2	2	2
		(ii)	$\% \text{ Unc in extension} = \frac{0.5 \times 100}{6} = 8.3 \text{ [\%]} (1)$ $\text{Total \% unc} = 8.3 \text{ \% (ecf if incorrect or omitted)} + 5 \text{ \%} + 11 \text{ \%} = 24.3 \text{ \% (1)}$ $\text{Absolute unc} = 130 \text{ (ecf)} \times \frac{24.3}{100}$ $= 3.083 \times 10^{10} (1)$ $= 31 \text{ G[Pa]} (1)$ N.B if 8 % used and 127 answer = 30 GPa If 8.3 % and 127 used answer = 31 GPa	1	1		4	3	4

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	(c)		Jack: The metal is made from copper (1) Karen: The metal could be zinc, bronze or copper (1) ecf must be consistent with answer to (b)			2	2		2
	(d)		Procedure – Jack may have: (award 2 marks max) • Taken repeat readings <u>of diameter</u> (1) • Measure mass more accurately (1) • Added weights in incremental steps / obtained <u>varying extensions</u> [with varying masses] / use a larger mass (1) • Use travelling microscope / vernier scale for extension (1) Don't accept different lengths Analysis – Jack may have: (award 1 mark max) • Draw a graph of force v extension or stress v strain [to find a mean value or gradient] (1)			3	3		3
			Question 4 total	4	5	5	14	8	14